

Automated Assessment of Residual Plots with Computer Vision Models

Response to reviewers for JCGS-25-535.R1

2026-04-14

We thank the reviewer for their time and constructive comments to improve this paper. The reviewer comments are colored in **purple** below and our response in black below. In addition, we also include an annotated version of the paper (`diff2.pdf`) with the difference between the original and revised versions produced using `latexdiff`.

Reviewer

1 Overall Comments

The idea of the paper is interesting and potentially useful. However, I think that the paper needs to be significantly revised to be solid. The paper contains two major ideas for residual diagnosis. One is the use of KL distance based measure D for discriminate good models and wrong models. The other idea is to use computer vision models to estimate D with $\hat{D}$. The methodology part of the paper needs to be separated clearly for these two tasks. The paper directly reports the performance of $\hat{D}$ but did not verify the goodness of D itself. *Although we see reasonable performance of $\hat{D}$, the goodness of D is still doubtful to me.* The measure D is a logarithm of a sum of KL distances of n observations. Therefore, the raw value of D is affected by the sample size n , the degree and types of model violation, and many other factors such as the number of predictors. Hence, the raw value of D may not be informative to detect model violation unless their distributions are calibrated by some methods. This may be why the authors use bootstrap to generate null and boot distributions of D to come up with a valid p-value. Unfortunately the description of the bootstrap procedure is unclear. It is suggested that the authors report the performance of the test p-value based on original D along with computer-vision-based $\hat{D}$, and describe the bootstrap procedure more clearly.

[TODO]

Thank you for your valuable comments.

2 Technical Critiques and Queries

pg 8, line 29:

The authors use the word “certain”. This makes it sound as though it is unknown. It is well known that in OLS, Q follows a rank-deficient multivariate normal, and $\|\hat{e}\|^2$ follows χ^2 with $df = n - p$.

We believe this is referring to the sentence “It follows that the distribution of residuals $\hat{e}$ can be characterized by a certain probability distribution, denoted as Q , which is transformed from the multivariate normal distribution.” [TODO] We thank the reviewer for pointing out the ambiguity in the wording. We have revised the sentence to clarify that Q is a specific distribution derived from the multivariate normal distribution of the residuals in OLS, and we have provided more details about its properties in the revised paper.

pg 10, eq 3 and eq 2:

Can the authors explain why the specific form $\log(1 + D_{KL})$ is used? I don’t see that eq 3 explains this form. Why +1 and why log?

The form $\log(1 + D_{KL})$ is used to ensure that large values of D_{KL} are compressed and to avoid taking the logarithm of zero when D_{KL} is zero.

About the D :

The features of D are unclear to me. Does its distribution depend on sample size and model violation types (for example, non-normality, omission of predictors, dependency)? Do the proposed tests try to get rid of this complicated distribution by using bootstrapping? I suspect that the D or $\hat{D}$ strongly depends on sample, as it is a sum of n terms. How can the D itself be useful for detecting model deviation?

[TODO]

pg 13: line 57: “we can apply bootstrapping to the estimated distance D ”:

The bootstrapping procedure is unclear. How is the bootstrapping for residuals performed? To detect model deviation, we need to have $\hat{D}$ from the null distribution. Do the authors generate datasets from the fitted model for a given dataset? I raised this question for the authors’ R0 version. The response letter said, “Specifically, assuming the fitted regression model is correct, we generate null residuals using residual rotation”. *How can we assume the fitted model is correct? How can a rotation of residuals from a wrong model produce residuals from fitting a true distribution?* In addition, the phrases “true residual plot” and “null residual plot” are used often by the authors. These phrases are technically confusing. A more detailed explanation is needed.

[TODO]

pg 14: line 42: “The model begins with an input layer of shape $n \times h \times w \times 3$, capable of handling n RGB images”.

What do n , h , w , and 3 mean? Why do the authors use RGB images for computer vision? Is there color in a residual plot?

The n is the sample size, h and w are the height and width of the image, and 3 is the number of color channels for RGB images. We use RGB images for computer vision because many pre-trained models are designed to work with RGB images, and it allows us to leverage transfer learning from models trained on large datasets. While residual plots typically do not contain color information, we can still represent them as RGB images by replicating the grayscale values across the three color channels. In our residual plot, the x axis is colored in red. This way, we can utilize the powerful feature extraction capabilities of pre-trained computer vision models. We have clarified this in the revised paper.

[TODO]

pg 16, Sections 4, 6, 7:

These sections are all about how to generate training datasets with known D to train the computer vision model f_{CV} . I think these three sections should be merged together into a single one. Section 5 should be moved after or before this section to explain how to use D or $\hat{D}$ for obtain a test p-value.

[TODO]

Sec 8.2.1:

The comparison with human visual inference is hard for me to follow. Who are the humans who made the visual decision? Are the participants experienced in looking at residual plots? Also, what's the difference between a "true residual plot" and a "null residual plot"? It seems to me that human judgment is subjective. It would be OK not to compare with human visual decision at all.

[TODO]

Fig 8: "Finally, the power curves in Figure 8 are fitted using logistic regression without intercepts and with an offset of $\log(0.05/0.95)$ to ensure a 5% rejection rate under H_0 ."

This way to report rejection rate is very wierd to me. Why don't the authors simply report the rejection rate, ie, proportion of rejections with $p < 0.05$, but fitting a model to the decisions. In addition, it seems that this is based on true D not $\hat{D}$. I think it is necessary to report the direct rejection rates of $\hat{D}$ and D . This may be the most convincing way to show the performance of the p-value based on D and $\hat{D}$.

[TODO]